

Karen Shieh

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EDUCATION

University of California, Berkeley

Berkeley, CA

B.A. in Computer Science, B.A. in Data Science | R&C Scholar | 4.00 GPA

Aug. 2022 – May 2026

- Related coursework: The Structure and Interpretation of Computer Programs, Data Structures, Discrete Mathematics and Probability Theory, Principles & Techniques of Data Science, Efficient Algorithms and Intractable Problems, Machine Structures, Introduction to Machine Learning, Optimization Models in Engineering, Computer Security, Designing/Visualizing/Understanding Deep Neural Networks, Operating Systems and System Programming, Natural Language Processing, Intro to Computer Vision and Computational Photography
- Awards and Honors: Regents' and Chancellor's Scholar, EECS Honors

EXPERIENCE

Undergraduate Student Researcher

Jan. 2025 – Present

University of California, Berkeley (REDS)

Berkeley, CA

- Developed an LLM-driven pipeline that automatically scrapes seed files from the web (e.g. mining GitHub repositories), designed as an off-the-shelf tool to lower the entry barrier to fuzzing
- Built Docker-based fuzzing infrastructure for reproducible experiments across environments
- Demonstrated that the generated corpus performs on par with expert-created corpora through Magma benchmark analysis, measuring vulnerability reach/trigger times and code coverage

Software Engineering Intern

May 2025 – August 2025

ClickTime

San Francisco, CA

- Developed C#/.NET backend APIs, created SQL tables with triggers, and built React frontend components for Time Entry Activity, a heatmap dashboard to visualize employee time entry behavior
- Designed a reusable and scalable full-stack architecture to ensure maintainability and support future extensions
- Built user-centric features that streamlined workflows and integrated with existing systems
- Leveraged GitHub Copilot to accelerate development efficiency and achieve stretch goals beyond the original project scope

Web Development Assistant

May 2024 – May 2025

Safe Transportation Research and Education Center

Berkeley, CA

- Built a dynamic SvelteKit website for Southern California cities to collect and manage data on bicycle, pedestrian, and wheelchair traffic for infrastructure projects
- Designed and maintained relationships between tables in the database through Prisma and SQL commands

PROJECTS

Estuary

Aug. 2024 – May 2025

- Working on an open-source multimodal framework for building low-latency real-time socially interactive agents as the front-end developer ([GitHub](#))
- Creating a realistic conversation environment between the agent and the user through Unity and Apple Vision Pro

NASA Spacesuit User Interface Technologies for Students Challenge

Nov. 2022 – Aug. 2023

- Collaborated with USC to design and develop spacesuit information displays and a navigation system for astronauts using HoloLens 2 during EVA and UIA
- Integrated AI, text-to-speech (TTS), and computer vision (CV) technologies into Unity, enhancing the user interface and experience
- Worked on the UI/UX design to ensure a seamless user experience in extreme environments, focusing on real-time feedback and intuitive controls

TECHNICAL SKILLS

Languages: Java, Python, C/C#, SQL, Scheme, JavaScript, TypeScript, HTML/CSS

Frameworks: React, Svelte/kit, Unity Game Engine, Jupyter Notebook, HoloLens 2, Qualtrics, Prisma, .NET

Developer Tools: Git, Docker, VS Code / Copilot, Visual Studio, PyCharm, Cursor, Google Colab, SSMS, Postman

Libraries: pandas, NumPy, Matplotlib, seaborn, PyTorch

Certifications: Cloud Computer Basics, Introduction to Programming with MATLAB, New Models of Business in Society, CAD Mechanical Design I, Computer Essential Skills & Programming